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Abstract: The extraction of traditional Chinese medicine (TCM) is a crucial step in the modernization of TCM, where the efficiency and energy consumption are significantly affected by the microscopic properties of the solvent system, such as polarity and hydrogen bonding capacity. Current optimization of extraction processes largely relies on empirical trial-and-error approaches, lacking a systematic understanding of the intrinsic relationships between solvent systems and the thermodynamic behavior and energy dynamics of the extraction process. Although existing studies have applied models like Apelblat and Jouyban-Acree to fit solubility data, they face limitations in accurately characterizing interactions in multicomponent solvent systems, clarifying the separation of thermodynamic parameters, and linking energy changes with underlying molecular mechanisms. To address these challenges, this study investigates the mechanistic influence of solvent systems on the thermodynamics and energy transformations during TCM extraction from two perspectives: (1) constructing thermodynamic models by quantifying the temperature-solubility relationship via the Apelblat model, analyzing synergistic effects of mixed solvents using the Jouyban-Acree model, and fitting solubility behavior in complex solvent systems through an integrated modeling approach; and (2) introducing the KAT-LSER model, which employs Kamlet-Taft solvent parameters to establish linear correlations between solvent molecular characteristics and changes in enthalpy and entropy, thereby isolating the contributions of hydrogen bonding, dipolar interactions, and other forces to the thermodynamic driving forces of dissolution. This research reveals how the microscopic properties of solvents regulate the enthalpy-entropy balance of dissolution, affecting the spontaneity and energy requirements of the extraction process. A quantitative framework linking "solvent composition–thermodynamic parameters–energy changes" is thus established. The results

provide theoretical support for a "thermodynamics-guided" optimization of TCM extraction, offering new insights into improving extraction efficiency, reducing energy consumption, and advancing green extraction technologies.

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